Docklands Community App
PROPOSAL

Introduction

This project is based on the Smart Docklands theme and is focused on improving accessibility and people's mobility. The goal of this app is to help people who are commuting or living in the Docklands area to be aware of the urban issues that can affect daily mobility or safety by creating a community system where users can check and report the status of these issues.

Problem Description

Docklands is a mix of residential, commercial and leisure areas which makes this place be daily visited by many people. As a result, people can often encounter issues that can affect their mobility and their safety, issues like closed walkways, damaged infrastructure, construction barriers. These problems are sometimes temporary but can still have a significant impact specially for people with mobility challenges.

Proposed Solution

The solution to this problem is an application called Docklands Community App. This application allows users to report issues in the Docklands area. The report includes information like location, description and severity. Once a report is created, it becomes visible for all users so they can be aware of the problem before reaching the location, helping them to make better decisions about their routes.

Smart Docklands Initiative

The Docklands Community App fits the goals of the Smart Docklands initiative by encouraging users to collaborate and share information about issues they encounter in the Docklands area. As a result, the application helps users better navigate the area and improves their safety.

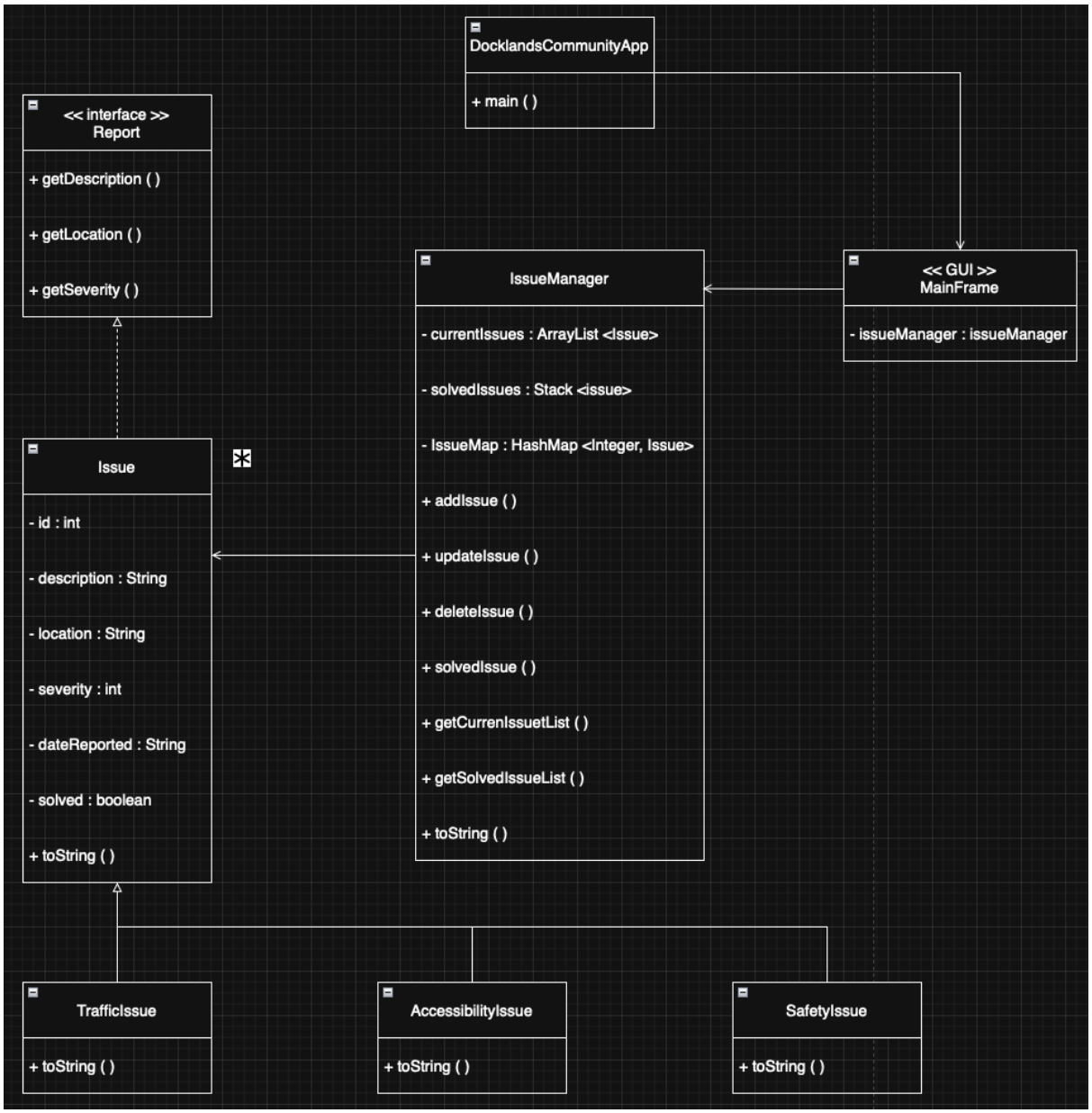
System Overview

The application simply consists of reporting issues in the Docklands area. The reports are filled with information provided by the user such as location, description and severity and the date the issue was reported.

All reports are stored and displayed in a list of current issues, which are used for users to check their routes and plan it accordingly. Allowing users to stay safe and informed before they even get to their destination.

Once the problem is no longer present, users can mark the option issue solved which will move the issue from "current issue list" to a different one called "solved issue list". Allowing users to see all issues solved, creating a history of reported problems.

UML Diagram


DocklandsCommunityApp
+ main ( )
<< interface >>
Report
+ getDescription ( )
+ getLocation ( )
+ getSeverity ( )
IssueManager
- currentIssues : ArrayList <Issue>
- solvedIssues : Stack <issue>
- IssueMap : HashMap <Integer, Issue>
+ addIssue ( )
+ updateIssue ( )
+ deleteIssue ( )
+ solvedIssue ( )
+ getCurrenIssuetList ( )
+ getSolvedIssueList ( )
+ toString ( )
<< GUI >>
MainFrame
- issueManager : issueManager
Issue
- id : int
- description : String
- location : String
- severity : int
- dateReported : String
- solved : boolean
+ toString ( )
TrafficIssue
+ toString ( )
AccessibilityIssue
+ toString ( )
SafetyIssue
+ toString ( )